Ram Villa

Senior Full-Stack Engineer
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PROFESSIONAL SUMMARY

Senior full-stack engineer with 6+ years building production web applications in regulated environments, across healthcare and fintech. Builds React and TypeScript frontends on serverless AWS backends (Lambda, Glue, DynamoDB, EventBridge), and designs releases that can be rolled back safely: blue-green data cutovers and staged feature-flag rollouts. Recently built an LLM chatbot on AWS Bedrock and AI developer tooling used in daily development. Builds to AODA and WCAG accessibility standards and works with health data under PHIPA.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java
Frontend: React, Angular, Redux, React Native, React Hook Form, Zod, Vite, Bootstrap
Backend and Data: Node.js, Express, REST APIs, WebSocket, PostgreSQL, DynamoDB, Redis
AWS (Compute and Messaging): Lambda, API Gateway, EventBridge, SQS, SNS
AWS (Data and Platform): S3, Glue, DynamoDB, SSM Parameter Store, Secrets Manager, CloudFront, SAM, Bedrock
DevOps and Testing: Docker, Jenkins, GitHub Actions, Git, Bitbucket, Jest, Cypress
AI Tooling: RAG, MCP, Claude Code, GitHub Copilot agents
Standards: AODA and WCAG accessibility, i18n, FHIR, OAuth 2.0, OIDC, PHIPA

PROFESSIONAL EXPERIENCE

Ontario Health - Ottawa, ON — Feb 2025 - Present
Senior Software Engineer (Full-Stack)

- Built the ONE Health chatbot with RAG over Ontario Health docs, using Claude on AWS Bedrock.
- Kept chatbot answers grounded in approved sources only, so responses stay within published Ontario Health documentation.
- Co-designed the S3, EventBridge, SQS, Lambda, and Glue ingestion pipeline into versioned DynamoDB tables, defined in AWS SAM.
- Ran blue-green DynamoDB cutovers via SSM Parameter Store pointers, making rollback a config change.
- Staged feature rollouts at 10/50/100% behind feature flags, limiting the blast radius of a bad release.
- Migrated multiple workspaces from Angular to React incrementally, shipping sprint by sprint.
- Kept Angular and React running side by side during the migration, with no feature freeze.
- Built a multi-step registration wizard with React Hook Form and Zod validation.
- Built the wizard to AODA and WCAG accessibility standards, with full i18n support.
- Delivered Canada Post address autocomplete end to end.
- Built the autocomplete as a debounced React search backed by a Lambda lookup.
- Cut provider-search latency from 850ms to 120ms with composite PostgreSQL indexes on a table of about 500K rows.
- Shrank the frontend bundle from 2.1MB to 1.3MB.
- Created an audit-logging npm package used in all three monorepo workspaces.
- Wrote Jest unit tests and Cypress end-to-end tests for the workspaces.
- Worked on FHIR, OAuth 2.0, and OIDC integrations for health-data exchange and sign-in.
- Fixed a privilege-escalation bug in a SMART on FHIR broker with OAuth 2.0 scope-based authorization.

Independent Projects - Ottawa, ON — Jun 2024 - Jan 2025
Full-Stack Developer

- Built PDF Chat, an app for asking questions over PDF documents, in TypeScript and React.
- Built Screenshot to Code, which turns UI screenshots into front-end code.
- Built a real-time whiteboard that syncs drawing live across connected users.
- Built a task CLI in Node.js for managing to-dos from the terminal.

University of Ottawa - Ottawa, ON — May 2023 - Apr 2024
Software Engineer (Co-op)

- Built React features with Hooks and Redux for a healthcare app, handling patient data under PHIPA.
- Shipped changes through Jenkins and GitHub Actions pipelines for testing and deployment.
- Worked with product managers and QA to prioritize features and clear dependencies.

Ionixx Technologies - India — Dec 2021 - Aug 2022
Senior Programmer

- Wrote React and React Native screens for Bitsian, a multi-exchange crypto trading platform.

- Built Node.js APIs for the Bitsian platform.
- Built frontend screens for Voyaj, a community travel app.
- Built API endpoints for Voyaj's community features.
- Implemented JWT and OAuth authentication to protect users' financial data.
- Containerized services with Docker during a move to microservices, and ran UI A/B tests.

Sims Healthcare Pvt Ltd - Hyderabad, India Jul 2018 - Dec 2021

Junior Software Developer

- Built an IoT patient-monitoring system with React Native, Node.js, and PostgreSQL.
- Built 12-lead ECG waveform streaming.
- Integrated medical devices with the monitoring platform.
- Added real-time provider-patient communication over WebSocket.
- Wrote secure REST APIs with encryption for patient data.
- Gathered requirements with clinicians for the monitoring system.
- Containerized services with Docker to keep dev and prod environments at parity.

PROJECTS

AI Developer Tooling | MCP, Claude Code, GitHub Copilot (Ontario Health)

- Built DocSync, a GitHub Copilot agent for the One Onboarding monorepo (Angular 19 and React), which opens draft doc-update PRs from merged diffs.
- DocSync classifies each merged change as an addition, modification, or removal, and drafts the matching documentation update.
- Fixed DocSync false negatives - a renamed session key, a removed helper - with a prompt rewrite using explicit rules and HALT/SKIP reporting.
- Replaced single-agent code search with five specialized Copilot agents - router, tracer, type hunter, cross-project, and synthesizer.
- Worked around cross-repo token limits with a repository_dispatch pattern and a scoped token.
- Built MCP servers for GitHub and AWS, used daily for repository and cloud work.
- Built Claude Code skills for gated plan, code, and audit reviews; consolidated 22 skills to the 6 worth keeping.

Jarvis (in progress) | GitHub Copilot custom agents

- Building a three-agent pipeline that turns a Jira ticket into a plan, code, and a PR, with approval at every step.
- Human approval gates at the plan, code, and PR steps keep a person in control of every change.

AI Trading Simulator (in progress) | React, Vite, Express, PostgreSQL, Claude Haiku

- Building an educational paper-trading app on Alpaca market data, with AI news analysis.
- Paper trading only, with real market data - no real funds at risk.

SpendVoice (in progress) | React PWA, Claude Haiku

- Building a voice-driven expense tracker that parses spoken entries.
- Projects 10-year savings from tracked spending.

EDUCATION

Master of Engineering - University of Ottawa, Ottawa, ON Sep 2022 - Dec 2023

Bachelor of Technology - JNTU, Hyderabad, India Sep 2014 - May 2018

CERTIFICATIONS

AWS Certified Developer - Associate